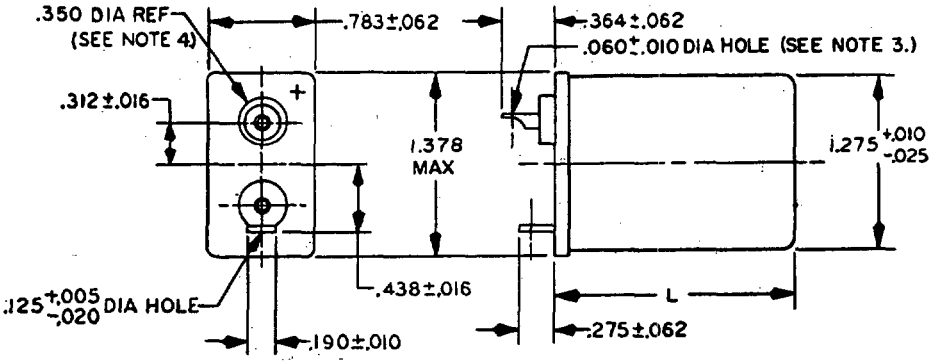
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM (POLARIZED, PLAIN FOIL), 85°C (VOLTAGE DERATED TO 125°C), STYLE CL51

INACTIVE FOR NEW EQUIPMENT DESIGN. NO SUPERSEDING DOCUMENT.

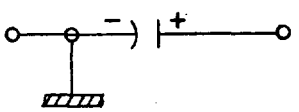
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

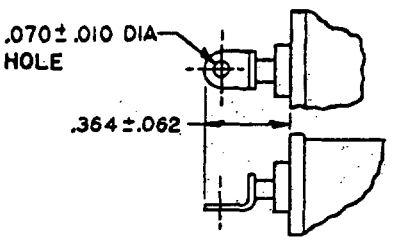


Dimensions	
	L ±.062 (1.57)
D1	1.375 (34.93)
D2	2.000 (50.80)
D3	2.500 (63.50)

TYPE D CASE



CIRCUIT DIAGRAM



OPTIONAL ALTERNATE
TERMINAL CONSTRUCTION

INCHES	MM	INCHES	MM
.005	.13	.190	4.83
.010	.25	.275	6.99
.016	.41	.312	7.92
.020	.51	.350	8.89
.025	.64	.364	9.25
.060	1.52	.438	11.13
.062	1.57	.783	19.89
.070	1.78	1.275	32.39
.125	3.18	1.378	35.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Orientation of terminal hole not fixed.
4. Glass to metal (hermetic) seal.

Ⓗ FIGURE 1. Style CL51 capacitors.

Ⓗ denotes changes

FSC 5910

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Ⓗ TABLE I. Capacitor characteristics.

Type designation	Voltage			Capacitance	DC leakage (max)			Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C	125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	µA	ohms	
CL51BH361MPG	30	20	34.5	360	50	400	500	7.8	D2
CL51BJ351MPG	50	30	57.5	350	73	585	730	8.0	D3
CL51BL750MPG	75	50	86.2	75	26	210	260	37.0	D1
CL51BL151MPG	75	50	86.2	150	52	415	520	18.5	D2
CL51BL221MPG	75	50	86.2	220	77	615	770	12.5	D3
CL51BI101MPG	112	75	129.0	100	52	415	520	29.5	D2
CL51BQ400MPG	150	100	172.0	40	28	225	280	70.0	D1
CL51BQ800MPG	150	100	172.0	80	56	450	560	35.0	D2
CL51BQ101MPG	150	100	172.0	100	70	560	700	30.0	D3

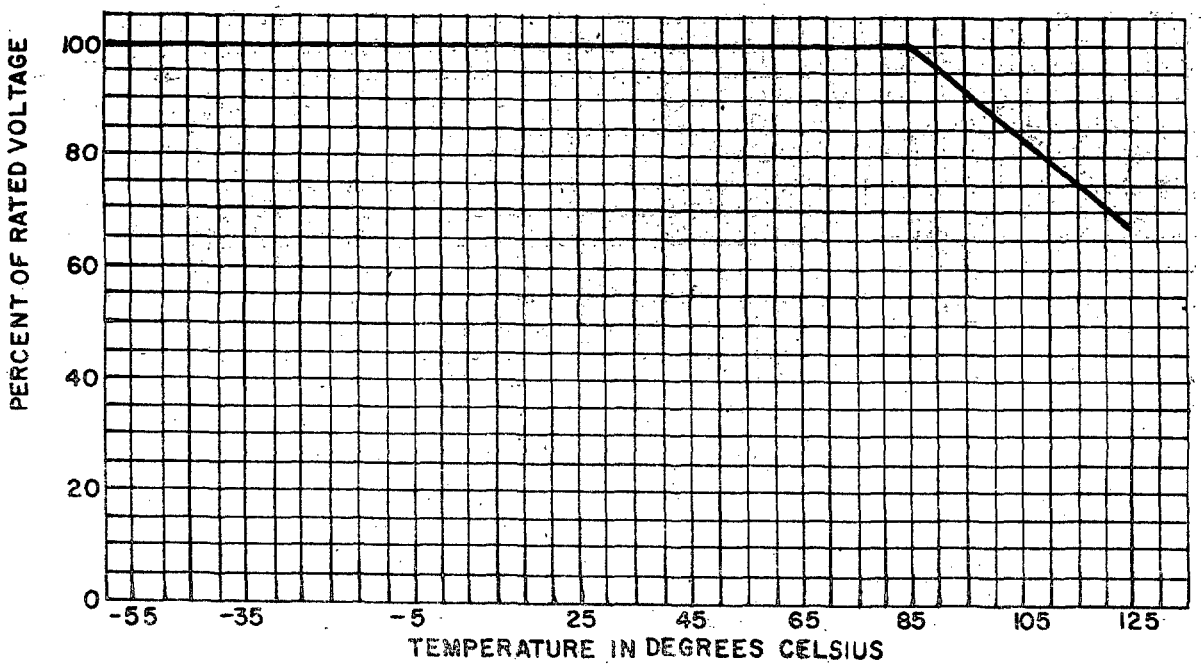


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration -
Capacitors - See figure 1.
Brackets - See figure 3.
Case type - Rectangular, uninsulated.
Seal type - Hermetic (symbol G).
Terminals - See figure 1.
Rated temperature - -55° C to +85° C, voltage derated to +125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol M (±20%).

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max): 10%.

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - Not applicable.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min) - Not applicable.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: Not applicable.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test (if req.) - In accordance with MIL-C-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±3% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap - Within ±3% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55° C):
Impedance - See table I.
ΔCap - Within -35% of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±3% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - 25%.
Step 5 (+125° C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - 25%.
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±3% of initial measurement.
DF (max) - Shall not exceed initial requirement.

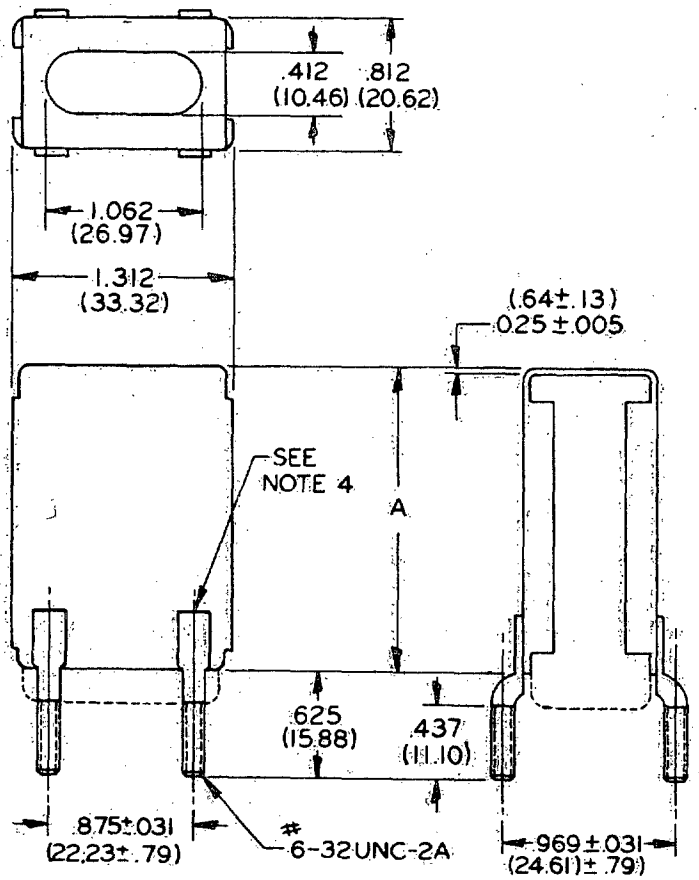
Reverse voltage: Not applicable.

Life at +85° C:
2,000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.
1,760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure: Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap (max) - +25% of initial measurement.
DF (max) - 150% of initial requirement

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BRACKETS FOR TYPE D CASE

Type designation	Case size	Dimensions						
		Dim	Inches			Millimeters		
			Min	Nom	Max	Min	Max	
CL050SD13	D1	A	1.159	1.175	1.191	29.44	30.25	
CL050SD23	D2	A	1.784	1.800	1.816	45.31	46.13	
CL050SD33	D3	A	2.284	2.300	2.316	58.01	58.82	

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 3. CL050 bracket.

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Marking: Capacitors shall be marked as shown in the following example:

Example:

CL51BE331MPG - Type designation.
12345 7015 - Source and date code.
330 μF 15 vdc - Capacitance and rated voltage (at 85° C).

Polarity-Plug (+) symbol near positive terminal (see figure 1).

Voltage groups (85° C rated voltage):
I - - - - - - - - - - - 15 to 50 v, incl.
II - - - - - - - - - - - 75 to 150 v, incl.

Substitution data: Capacitors covered by this specification are directly substitutable for capacitors of equivalent capacitance, capacitance tolerance, and voltage in the same case size.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 85

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